overcorrection. Rather, this is simply the expected behavior of a spread based on relatively straightforward dynamics for the two variables that comprise the spread – i.e. a mean-reverting variable with a half-life of one month attracted to a mean-reverting variable with a half-life of three months.[4]

Another way to represent the dynamics introduced by the drift coefficient in the stochastic differential equation is to consider the vector field introduced by the transition matrix, shown as Figure 4.7.

The horizontal axis corresponds to the value of the first variable in this example, while the vertical axis corresponds to the value of the second variable, such that the values of the two variables at any point in time can be represented by a point on this graph.

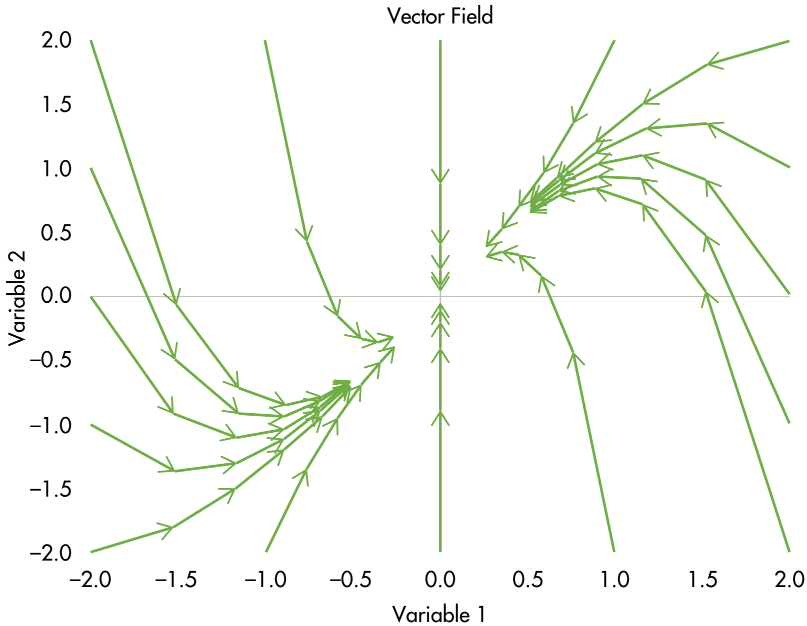

FIGURE 4.7 Vector field corresponding to the transition matrix.
Source: Authors.

[4]Apparent "overshooting" of the sort shown in Figure 4.6 is sometimes attributed to momentum exhibited by a series. While we do believe there are times when genuine momentum is present in a data set, this example illustrates that 'overshooting' can be exhibited by linear systems even in the absence of momentum.

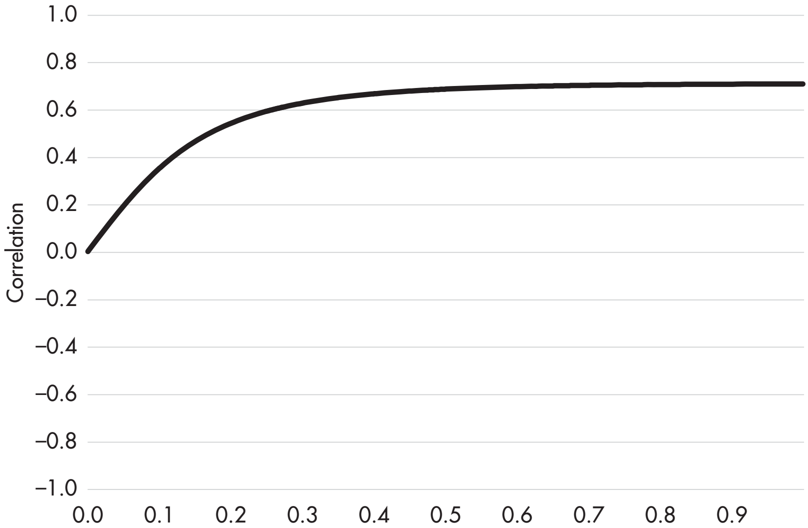

FIGURE 4.8 Correlation as a function of horizon.
Source: Authors.

In the absence of any stochastic effects introduced by the Brownian motions, the two variables would follow the arrows that illustrate the vector field in Figure 4.7. The curvature apparent in this vector field representation is due to the fact that the second variable is attracted by the first.

Also interesting in this example is the correlation between the two variables as a function of horizon, shown in Figure 4.8.

The correlation between the two variables over a horizon of one day is nearly zero, as we've specified that the two variables are not affected by common shocks, and one day isn't much time for the attraction of one variable to the other to affect the correlation. But on a horizon of one month, the correlation has increased to 0.30, and on a horizon of three months, the correlation has increased to 0.6, owing to the cumulative effect of the second variable being attracted to the first variable over time.

Next, let's leave the transition matrix the same as in the last example, but let's change the scatter matrix so that the two variables react in different directions to changes in the two Brownian motions. In other words, let's specify the scatter matrix

$$\boldsymbol{S} = \begin{bmatrix} 1 & -0.2 \\ -0.2 & 1 \end{bmatrix}$$

Our intuition is that the correlation should be negative over short horizons, owing to the fact that the two variables react in opposite directions to

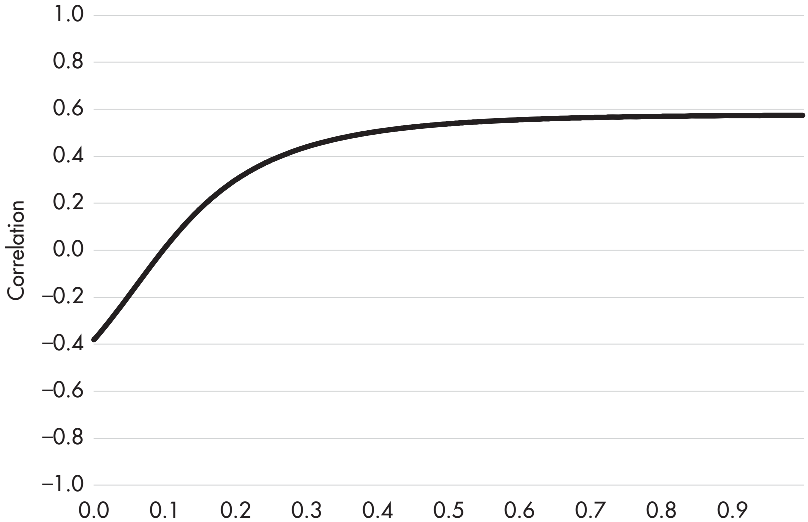

FIGURE 4.9 Correlation as a function of horizon with opposite reactions to changes in Brownian motions.
Source: Authors.

changes in the two Brownian motions. But we should also expect this negative correlation to lessen over time, as the effect of the second variable being attracted to the first has a chance to affect the correlation. Figure 4.9 shows this correlation as a function of the horizon.

In this case, the correlation not only lessens over time; it changes sign. The correlation is -0.4 over a one-day horizon but is 0.4 on a three-month horizon.

Horizon-dependent correlations would have significant implications for portfolio construction in general and for relative value trading in particular, if we observed them in practice. And, as it happens, we do observe them in practice – fairly regularly, in fact. As a result, we suggest relative value analysts calculate correlations over multiple horizons (e.g. one day, one week, two weeks, three weeks, one month, two months, three months). If we observe horizon-dependent correlations in our data, we probably should model the data with a process that admits horizon-dependent correlations. And given its flexibility, the MVOU is a strong candidate in such cases.

Conditional Density

The conditional density of the MVOU process is multivariate normal with mean vector equal to

$$E[\boldsymbol{X}_t] = \boldsymbol{\mu} + e^{-\boldsymbol{\Theta}(t-s)}(\boldsymbol{X}_s - \boldsymbol{\mu}), \text{for } t > s,$$

and covariance matrix

$$vec(\boldsymbol{\Sigma}_t) = (\boldsymbol{\Theta} \oplus \boldsymbol{\Theta})^{-1}(\boldsymbol{I}_{m^2} - e^{-(\boldsymbol{\Theta}\oplus\boldsymbol{\Theta})(t-s)})vec(\boldsymbol{\Sigma})$$

where $\Sigma = \boldsymbol{SS}'$ and $\boldsymbol{I}_{m^2}$ is the m^2 x m^2 identity matrix.

The text box contains some tips for calculating these quantities.

Calculating the Conditional Covariance Matrix

THE *vec*() FUNCTION

The *vec*() function simply flattens an array. For example,

$$vec\begin{bmatrix} a & c \\ b & d \end{bmatrix} = [a\ b\ c\ d].$$

We'll use the notation, *inversevec*[], to refer to the inverse of this flattening.

$$inversevec[a\ b\ c\ d] = \begin{bmatrix} a & c \\ b & d \end{bmatrix}.$$

THE KRONECKER PRODUCT

$\otimes$ denotes the Kronecker product. If $\boldsymbol{A}$ is an m x n matrix and $\boldsymbol{B}$ is a p x q matrix, then $\boldsymbol{A} \otimes \boldsymbol{B}$ is an mp x nq matrix of the form

$$\begin{bmatrix} a_{11}\boldsymbol{B} & \cdots & a_{1n}\boldsymbol{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\boldsymbol{B} & \cdots & a_{mn}\boldsymbol{B} \end{bmatrix}.$$

THE KRONECKER SUM

$\oplus$ denotes the Kronecker sum. If $\boldsymbol{A}$ is an m x m matrix and $\boldsymbol{B}$ is an n x n matrix, then

$$\boldsymbol{A} \oplus \boldsymbol{B} = \boldsymbol{A} \otimes \boldsymbol{I}_n + \boldsymbol{B} \otimes \boldsymbol{I}_m$$

THE MATRIX EXPONENTIAL

There are a few ways to calculate the exponential of a matrix. It's worth stressing that simply calculating the exponential of each element in the matrix is ***not*** one of these ways.

(*continued*)

(*continued*)

In our experience, one of the most robust ways to calculate a matrix exponential is to do the following:

1. Decompose a matrix into its eigenvectors and eigenvalues.
2. Replace each eigenvalue with its exponential.
3. Reformulate the matrix as a product of its eigenvectors and the exponentiated eigenvalues.

For example, we can express the m x m matrix, $\boldsymbol{A}$, as $\boldsymbol{Q\Lambda Q}^{-1}$, where $\boldsymbol{Q}$ is an m x m matrix in which each column is an eigenvector of $\boldsymbol{A}$, and $\boldsymbol{\Lambda}$ is an m x m diagonal matrix, in which each element of the main diagonal is an eigenvalue of $\boldsymbol{A}$. Define the matrix, $\boldsymbol{\Omega}$, to be the diagonal matrix containing the exponentials of the eigenvalues in $\boldsymbol{\Lambda}$. In other words, with $\boldsymbol{\Lambda} = \begin{bmatrix} \boldsymbol{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \boldsymbol{\lambda_m} \end{bmatrix}$, we have

$$\boldsymbol{\Omega} = \begin{bmatrix} e^{\lambda_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & e^{\lambda_m} \end{bmatrix}$$

Then $e^{\boldsymbol{A}} = \boldsymbol{Q\Omega Q}^{-1}$

Unconditional Density

The unconditional density of the MVOU process is also multivariate normal, with a mean vector equal to the limit of the conditional mean vector as $t - s \rightarrow \infty$

$$\lim_{t-s\rightarrow\infty} [\boldsymbol{\mu} + e^{-\boldsymbol{\Theta}(t-s)}(\boldsymbol{X}_s - \boldsymbol{\mu})] = \boldsymbol{\mu}$$

And with a covariance matrix equal to the limit as $t - s \rightarrow \infty$ of the conditional covariance matrix

$$\lim_{t-s\rightarrow\infty} [(\boldsymbol{\Theta} \oplus \boldsymbol{\Theta})^{-1}(\boldsymbol{I}_{m^2} - e^{-(\boldsymbol{\Theta}\oplus\boldsymbol{\Theta})(t-s)})vec(\boldsymbol{\Sigma})] = (\boldsymbol{\Theta} \oplus \boldsymbol{\Theta})^{-1}vec(\boldsymbol{\Sigma})$$

So that the unconditional covariance matrix is

$$inversevec[(\boldsymbol{\Theta} \oplus \boldsymbol{\Theta})^{-1}vec(\boldsymbol{\Sigma})]$$

Likelihood Function and Log Likelihood Function

As usual, the likelihood function is the product of the conditional densities, and the log likelihood function is the sum of the natural logarithms of the conditional densities.

Some analysts make use of the first observation in a data set by adding to the likelihood and log likelihood functions the unconditional density applied to the first observation.

EXAMPLES

BTP Butterfly

One of our favorite applications of the MVOU model was the butterfly spread involving three BTPs: 1.65% of Mar-32, 2.25% Sep-36, and 5% Sep-40, discussed in Huggins (Jan 2019) and shown in Figure 4.10 from January 2017 through January 2019.

BPV-neutral butterfly spreads can have a tendency to be directional. In fact, we get some indication of this already from Figure 4.10, which shows that the yield of the Mar-32 issue increased by a greater amount than did the yields of the other issues in the sell-off of May of 2018. Figure 4.11 illustrates the apparent strength of the directionality by showing the average of the three

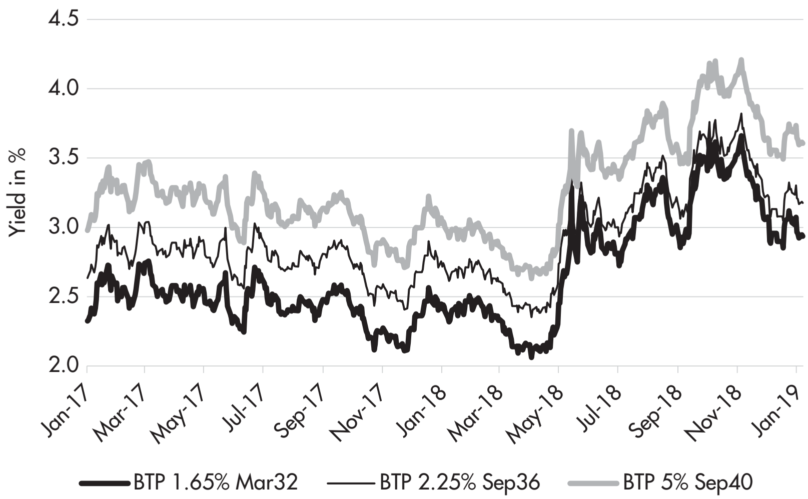

FIGURE 4.10 Yields of BTP 1.65% Mar-32, BTP 2.25% Sep-36, and BTP 5% Sep-40.
Source: Huggins (Jan 2019).

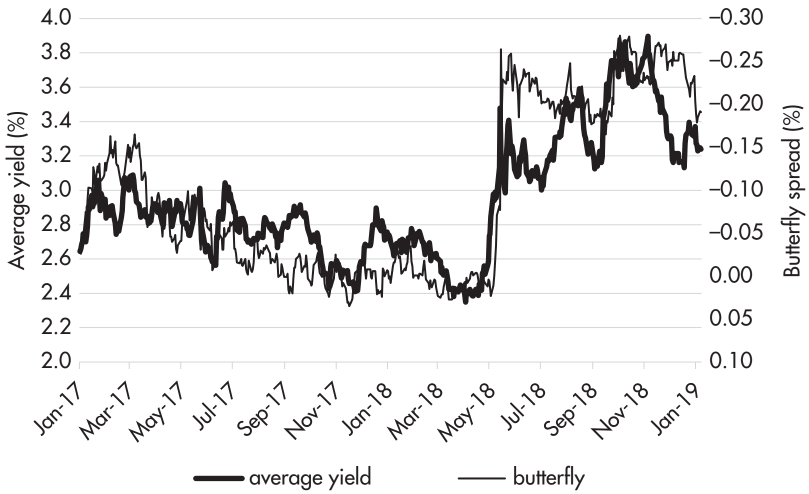

FIGURE 4.11 Butterfly spread and average yield.
Source: Huggins (Jan 2019).

bond yields along with the butterfly spread. We show the butterfly spread on an inverted axis to highlight the strong negative correlation between the spread and the yield level.

In fact, if we estimate the correlation coefficient between the average yield and the butterfly spread, without differencing either variable, we obtain an estimate of −0.88, further highlighting the strong directionality of this trade.

But many analysts prefer to compute instead the correlation coefficient of the *changes* in the variables. And as it happens, the estimated correlation coefficient between the daily change in the average yield and the daily change in the butterfly spread is 0.06. This positive correlation estimate appears quite counterintuitive, given the two series shown in Figure 4.11. In an attempt to reconcile our intuition with the calculated correlation coefficient, we might try using weekly data rather than daily data. And, in fact, this does give us a negative result of −0.27.

That number still seems a little low, given the two series shown in Figure 4.11, so we might recalculate the correlation coefficient using a data frequency of two weeks. And as it happens, the estimate is even more negative, at −0.57.

The good news is that our estimated correlation coefficient accords better with our intuition as we increase the time between successive observations. But the bad news is that it seems as if we're massaging the data quite a bit to get a number we like. By moving from a daily frequency to a frequency of two weeks, we've thrown out 90% of the data in our sample. As a general rule, if we have to throw out 90% of a data sample for our model to make sense, it's probably a good idea to reconsider our model.

So let's see how the MVOU model performs in this situation.

Initializing the Optimization Routine When Performing Maximum Likelihood

The three-variable system, modeled in the case of our BTP butterfly, isn't particularly large, but it does have 21 parameters we need to estimate: three long-run means, nine elements of the transition matrix, and nine elements of the scatter matrix.

With this number of parameters, it's useful to initialize the optimization algorithm at a point in the parameter space that is likely to be near (in some sense) the solution. One way to do this is to match empirical moments of the data.

For the three long-run means, the simple averages are usually very good starting points. Particularly when an analyst has an ample amount of data, these simple averages are usually quite close to the values for the long-run means that correspond to the optimized value of the likelihood function.

For the elements of the transition matrix, we suggest choosing elements of $\boldsymbol{\Theta}$ so that the elements of $e^{-\boldsymbol{\Theta}\tau}$ (for τ equal to one day) match as closely as possible the elements of the transpose of the linear regression coefficient matrix, $(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}\dot{\mathbf{X}}'\mathbf{X}$, where $\boldsymbol{X}$ is the $n\ x\ m$ matrix containing n observations of the m variables, and $\dot{\boldsymbol{X}}$ is the data matrix lagged by one day. In particular, we tend to choose the elements of $\boldsymbol{\Theta}$ to minimize the sum of squared differences between the elements of $e^{-\boldsymbol{\Theta}\tau}$ and the elements of $\boldsymbol{X}'\dot{\boldsymbol{X}}(\dot{\mathbf{X}}'\dot{\mathbf{X}})^{-1}$.

For the elements of the scatter matrix, we suggest choosing elements of $\boldsymbol{S}$ so that the elements of the one-day conditional covariance matrix in the model match as closely as possible the elements of the empirical covariance matrix using one-day changes in each data series. As with the transition matrix, we tend to choose these elements to minimize the sum of squared differences between these two matrices.

Figure 4.12 shows the estimated correlation coefficient as a function of the horizon.

As we discussed, the estimated correlation coefficient is clearly dependent on the horizon. No doubt the graph would look a little better if the estimate

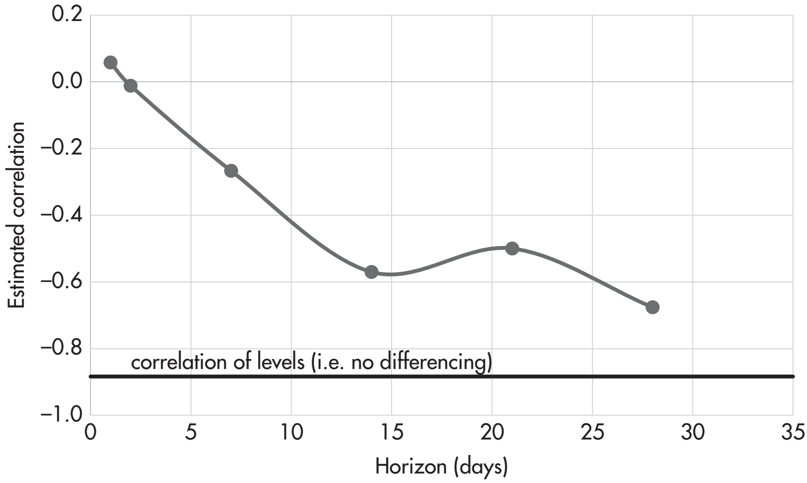

FIGURE 4.12 Correlation as a function of horizon.
Source: Authors.

of the correlation coefficient over the three-week horizon was between the estimate for the two-week horizon and the four-week horizon. But we only have 35 non-overlapping observations for the three-week horizon, so we need to make allowances for some degree of estimation error in these estimates. But in our view, the pattern is nearly a textbook example of horizon-dependent correlation that is well modeled by the MVOU process.

Figure 4.13 shows the empirical correlation coefficients shown above along with the fitted correlation coefficients produced by the MVOU model once calibrated to the data.

In our view, the fitted MVOU process does a pretty good job of modeling the horizon-dependent correlations between the butterfly spread and the average yield, capturing the positive correlation over short horizons and approaching the level correlation (i.e. the correlation of the data without differencing) asymptotically, coming fairly close to the estimated correlation coefficients for other horizons at the same time.

This calibration tells a story in which the three yields are highly correlated, even over a horizon of one day. But on any given day, the yield of one bond may move more or less than the yields of the other two bonds, in a manner that causes the butterfly spread to be essentially uncorrelated with the yield level *on that day*. But these somewhat random, idiosyncratic one-day changes to the butterfly spread tend to 'correct' over time owing to the attractions of the yields to one another.

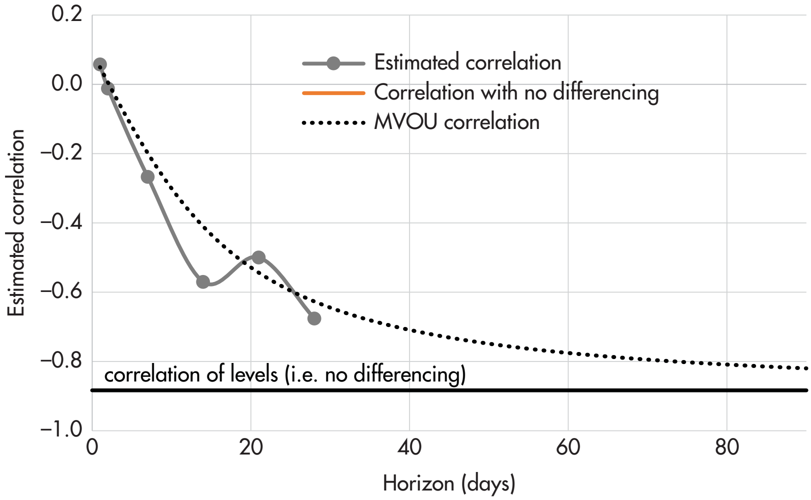

FIGURE 4.13 Correlation coefficient: estimated from data and fitted by MVOU model.
Source: Authors.

So with all this in mind, how are we to answer the question regarding the directionality of the butterfly spread? Is selling the Sep-36 issue against the Mar-32 and Sep-40 issues just a proxy for being long the BTP market? The answer is that it depends on the horizon. Over short horizons, there is essentially no directionality in the butterfly spread. But over longer horizons, the butterfly exhibits considerable directionality.

A trader might take the view that the horizon over which he expects the trade to perform is greater than one month, so that the relevant correlation to use is a negative number in the vicinity of, say, -0.70. But if he were to rely on that negative correlation when constructing a portfolio of trades, he's likely to find that his portfolio exhibits greater volatility over shorter horizons than he expected, owing to the fact that, over short horizons, his butterfly spread wasn't exhibiting much directionality at all.

This is also a problem for risk managers, who typically don't incorporate horizon-dependent variances and covariances into their analysis. It's also likely to pose a problem for anyone calculating risk-adjusted returns for the portfolio, as the calculated ex post correlation between the butterfly spread and the direction of the market, and therefore the volatility of the portfolio, will depend on the data interval over which observations are taken. For example, a reported Sharpe ratio might look materially different using daily data than using monthly data.